

Public Economics

Lecture 2 · Theoretical Tools of Public Finance

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Winter Term

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Today's Questions

- How do individuals choose how much to consume and how hard to work?
- How do firms choose how much to produce?
- What is the theoretical effect of raising — or cutting — Austria's minimum-income support (**Sozialhilfe**) on economic efficiency?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 2; Stantcheva, S., “Theoretical Tools for Public Economics” (Harvard, EC2450A).

2.1 Constrained Utility Maximization

2.2 Putting the Tools to Work: Sozialhilfe and Labor Supply

2.3 Equilibrium and Social Welfare

2.4 Welfare Implications of Reforming Sozialhilfe

2.5 Conclusion

Theoretical vs. Empirical Tools

Theoretical tools

The set of tools designed to understand the *mechanics* behind economic decision making — graphical and mathematical models of utility maximization subject to constraints.

Empirical tools

The set of tools designed to *analyze data* and answer the questions raised by theoretical analysis (the subject of Lecture 3).

Theory tells us the *direction* of an effect; data tells us its *size*. We need both.

2.1 Constrained Utility Maximization

Utility function

A mathematical mapping of an individual's consumption choices into a level of well-being: $U = u(X_1, X_2, \dots)$.

- Consumers are assumed to undertake **constrained utility maximization**: maximize well-being subject to available resources.
- Key assumption: **non-satiation** (“more is better”) — more of a good is always at least as good as less.

Running example: Lena, a WU student, choosing between *coffees at the Kaffeehaus* (C) and *cinema visits* (M).

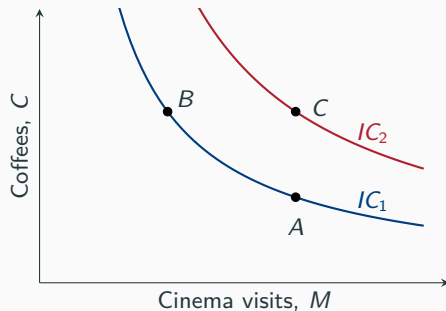
Indifference Curves

Indifference curve

All bundles of goods that leave an individual equally well off.

- Bundle *A*: 2 coffees, 1 cinema visit
- Bundle *B*: 1 coffee, 2 cinema visits
- Bundle *C*: 2 coffees, 2 cinema visits

Lena is indifferent between *A* and *B*, but strictly prefers *C* to both (more of everything).



Two properties (both from “more is better”): (1) higher indifference curves are preferred; (2) indifference curves always slope downward.

Marginal utility

The additional utility from consuming one more unit of a good: $MU_1 = \partial u / \partial X_1$.

- Example: $U = \sqrt{C \times M}$ (holding $C = 2$ fixed). Moving from 0 to 1 cinema visit raises U from 0 to $\sqrt{2} = 1.41$; from 1 to 2 visits, U rises only to $\sqrt{4} = 2$ (a gain of 0.59); from 2 to 3, the gain is only 0.45.
- This is **diminishing marginal utility**: each additional unit of a good raises utility by less than the previous unit.

Aside: Does Marginal Utility Always Diminish?

Not for addictive goods

Becker & Murphy (1988), “A Rational Theory of Addiction”: today’s utility from consuming a good can depend on *past* consumption of it — you build a habit.

- The more you have already consumed, the more you enjoy (or “need”) consuming more — the opposite of diminishing MU.
- This is a leading theoretical framework for taxing tobacco, alcohol, and sugar: if consumers systematically under-weigh future harm, corrective taxes can improve welfare beyond just correcting externalities.

Marginal Rate of Substitution

Marginal rate of substitution (MRS)

The rate at which a consumer is willing to trade one good for another; equals (minus) the slope of the indifference curve: $MRS_{C,M} = MU_M/MU_C$.

- For $U = \sqrt{C \times M}$: $MRS_{C,M} = C/M$.
- **Diminishing MRS**: as Lena has more cinema visits and fewer coffees, she is willing to give up less and less coffee for one more visit.

Diminishing MRS is exactly why indifference curves are *convex* (bowed toward the origin).

Budget constraint

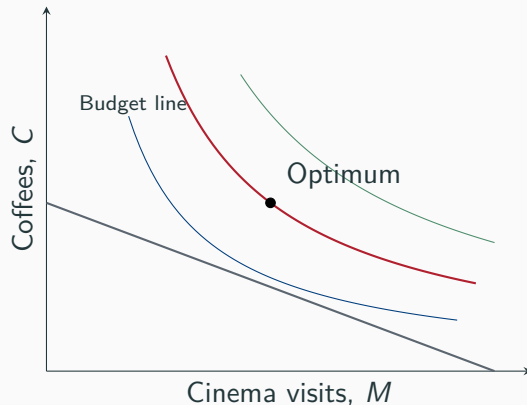
All combinations of goods an individual can afford, spending her entire income:

$$P_C C + P_M M = Y.$$

- Suppose a Kaffeehaus coffee costs €4, a cinema ticket costs €8, and Lena's monthly "fun budget" is €96.
- Maximum coffees (0 cinema): $96/4 = 24$. Maximum cinema visits (0 coffee): $96/8 = 12$.
- Slope of the budget line: $-P_M/P_C = -8/4 = -2$ — each extra cinema visit costs 2 coffees.

Constrained Choice: The Tangency Condition

- Lena's optimal bundle is where her highest reachable indifference curve is **tangent** to her budget line.
- At that point:
 $MRS_{C,M} = P_M/P_C$.
- With $U = \sqrt{C \times M}$ and this budget: optimum is $M = 6$, $C = 12$ (she splits her €96 evenly: €48 on each good).



Income and Substitution Effects

Suppose the price of cinema tickets rises. This has two distinct effects on demand:

Substitution effect

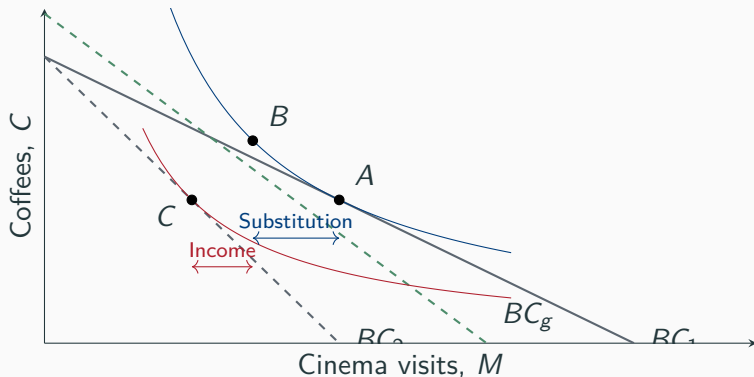
Holding utility constant, a relative price rise *always* causes less consumption of that good.

Income effect

A price rise makes the consumer effectively poorer, typically reducing consumption of *all* normal goods.

For a normal good, both effects push demand in the *same* direction — toward less of the good whose price rose.

Decomposing a Price Change



Cinema price rises: $A \rightarrow C$ overall. $A \rightarrow B$ (holding utility at IC_1) isolates the *substitution effect*; $B \rightarrow C$ isolates the *income effect*.

Normal vs. Inferior Goods

Normal good

Demand *increases* with income (most goods).

Inferior good

Demand *decreases* with income.

Austrian example

Public transport (*Öffis*) use tends to fall as household income rises — once people can afford a car, many drive instead. Public transport behaves like an **inferior good** over part of the income range, even though most goods are normal.

This is exactly why the KlimaTicket (Lecture 1) can only partly be understood through the price effect alone — income effects and preferences over cars vs. transit matter too.

2.2 Putting the Tools to Work: Sozialhilfe and Labor Supply

Austria's Sozialhilfe (Minimum Income Support)

- Since the 2019 *Sozialhilfe-Grundsatzgesetz*, Austria's means-tested minimum income sets **maximum** rates (*Höchstsätze*), not guaranteed minimums, with details varying by *Land*.
- Maximum monthly rate for a single adult/single parent (2024): $\approx$ €1,156. Average *actual* payment to single adults: $\approx$ €766/month.
- A 2019 reform introduced the *Wiedereinsteigerfreibetrag*: up to 35% of net earnings are disregarded (not counted against the benefit) for up to 12 months when a recipient starts working.

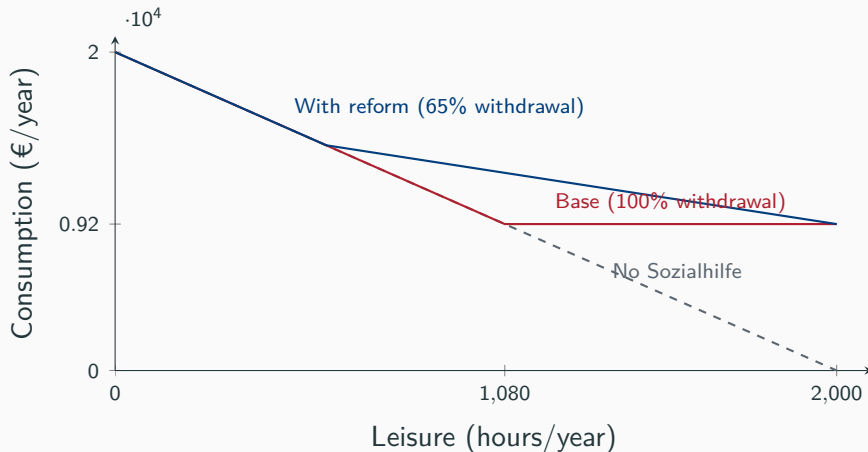
This is Austria's own version of Gruber's TANF example — a benefit guarantee plus a benefit-reduction rate. Source: ÖGB, “So viel Sozialhilfe bekommt man in Österreich”;

Sozialministerium, Sozialhilfe-Grundsatzgesetz.

The Consumption–Leisure Trade-off

- Suppose Julia, a single parent, can work up to 2,000 hours/year at a wage of €10/hour, with no other income and no Sozialhilfe.
- Each hour of leisure “costs” her €10 in forgone wages (opportunity cost).
- Her budget constraint: maximum €20,000/year in consumption (0 leisure) or 2,000 hours of leisure (€0 consumption); slope = -10 .

Adding Sozialhilfe to the Budget Constraint



Without any earnings disregard, Julia's net return to work is €0 until she earns her way off Sozialhilfe entirely — a pure “welfare trap.”

The Wiedereinsteigerfreibetrag: A Real Reduction-Rate Experiment

- **Base regime:** 100% benefit-reduction rate. Every euro earned reduces Sozialhilfe by one euro — net return to work is €0 (flat budget segment).
- **Reform regime:** 35% of net earnings disregarded $\Rightarrow$ effective reduction rate $\approx 65\%$. Net return to work rises to €3.50 per hour (slope -3.5 instead of flat).

Why this matters

This is *exactly* the benefit-guarantee/reduction-rate logic from Gruber's TANF example — applied to a real 2019 Austrian reform explicitly designed to soften the work disincentive.

Predicted Effects on Labor Supply

- Cutting the reduction rate (100% $\rightarrow$ 65%) makes work pay more — a pure **substitution effect** toward less leisure (more work).
- But it also raises Julia's income at any given hours choice — a competing **income effect** toward *more* leisure.
- The net effect depends on the relative strength of the two — and on how much Julia values leisure relative to consumption (her preferences).

Theory alone cannot say whether labor supply rises or by how much — exactly as in Gruber's Naomi/Sarah comparison.

How Large Will the Response Be?

- The constrained-choice model tells us the *direction* of well-behaved responses — but not their *magnitude*.
- Whether cutting Sozialhilfe's effective reduction rate actually raises single parents' labor supply, and by how much, is an empirical question.

This is exactly what Lecture 3 (empirical tools) is built to answer.

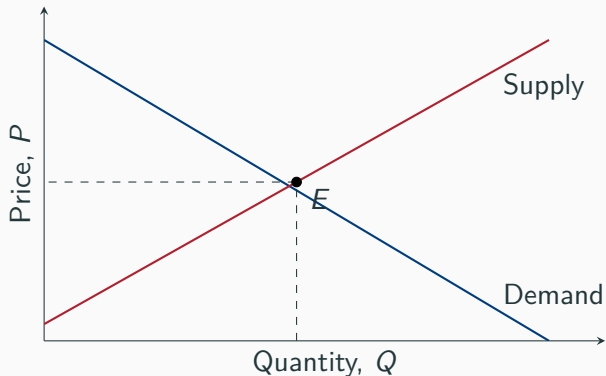
2.3 Equilibrium and Social Welfare

From Individual Choice to Market Demand

- Each individual's utility maximization generates her own demand $X(P, Y)$ for a good.
- The **market demand curve** is the horizontal sum of all individual demands at each price.
- **Elasticity of demand:** $\varepsilon_D = \frac{\% \Delta \text{quantity demanded}}{\% \Delta \text{price}}$ — typically negative, and typically not constant along the curve.

- Firms use a **production function** to turn inputs (labor, capital) into output, subject to **diminishing marginal productivity**.
- Profit maximization ($\max_Q [p \cdot Q - c(Q)]$) implies firms produce until **marginal cost equals price**.
- The marginal cost curve *is* the firm's supply curve; market supply is the horizontal sum across firms.

Market Equilibrium



Market equilibrium: the unique price/quantity pair at which demand equals supply, and both sides of the market are satisfied.

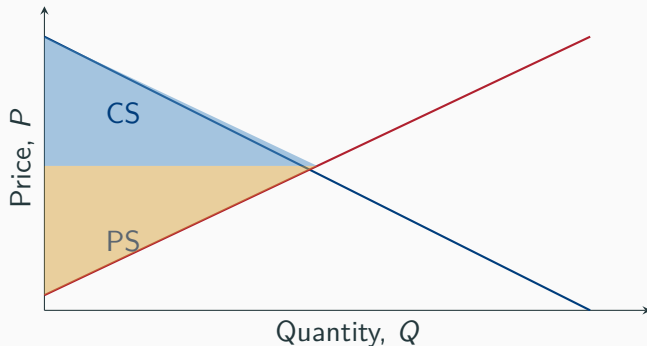
Consumer and Producer Surplus

Consumer surplus

The value consumers get from a good, above and beyond what they paid. Area below demand, above P_E .

Producer surplus

The profit producers earn, above and beyond their cost. Area above supply, below P_E .



The First Fundamental Theorem of Welfare Economics

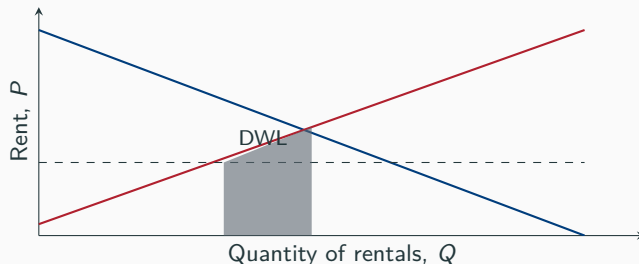
First Fundamental Theorem

The competitive equilibrium, where supply equals demand, maximizes total economic surplus (efficiency).

- Total surplus just counts € — it is blind to *who* gets them.
- Holds under: no externalities, perfect competition, perfect information, rational agents.
- When these assumptions fail (market failures), government intervention can potentially raise — not just redistribute — total surplus.

Application: Rent Regulation in Vienna

- About **21%** of Vienna's rented apartments are subject to the *Richtwertmietzins* price ceiling (currently €6.74/m²), below the market rent.
- A price ceiling below P_E reduces the quantity landlords are willing to supply — the classic price-ceiling case.



Source: Arbeiterkammer Wien, "Richtwertmietzins"; Stadt Wien, Mietenrechner.

The Second Fundamental Theorem of Welfare Economics

Second Fundamental Theorem

Any Pareto-efficient allocation can be reached by (1) suitably redistributing initial endowments via **lump-sum** transfers, then (2) letting markets work freely.

If this worked in practice, there would be *no conflict* between efficiency and equity.

Why the Second Theorem Fails in Practice

- Lump-sum redistribution requires the government to observe individual *characteristics* (need, ability) — not just *outcomes* (income, work status).
- In reality, the government usually cannot tell these apart, and must base taxes/transfers on observable behavior instead.

Illustration

Suppose 50% of the population is disabled and cannot work (income €0), 50% can work and earn €100. If the government could identify disability directly, it could tax the able €50 *regardless of whether they work*, and give €50 to each disabled person — full redistribution, zero distortion.

But if the government cannot tell a disabled person from an able person who simply isn't working, that same €50 tax-and-transfer destroys the incentive to work — redistribution now costs efficiency.

Social Welfare Functions

Social welfare function (SWF)

A function combining all individuals' utilities into one overall measure of social welfare.

Utilitarian

$$SWF = U_1 + U_2 + \cdots + U_N$$

Equal weight on every util. With diminishing marginal utility of income, favors *some* redistribution from rich to poor.

Rawlsian

$$SWF = \min(U_1, U_2, \dots, U_N)$$

Maximize the well-being of the *worst-off* member — generally implies *more* redistribution than utilitarianism.

Other Social Justice Principles

1. **Commodity egalitarianism:** society should guarantee basic needs (education, health care, retirement) as rights; beyond that, distribution doesn't matter.
2. **Equality of opportunity:** individuals should be compensated for circumstances beyond their control (family background, ability) — but not for the consequences of their own choices.
3. **Just deserts:** compensation should match contribution — taxes should track the benefits an individual receives from government (a libertarian view).

None of these reduce cleanly to utilitarian or Rawlsian social welfare — real policy debates mix all three.

What Do People Actually Think Is Fair?

Saez & Stantcheva (2016) surveyed $\approx 1,100$ US respondents with hypothetical vignettes: two people have identical pre-tax and post-tax income — who deserves a €1,000 tax break more?

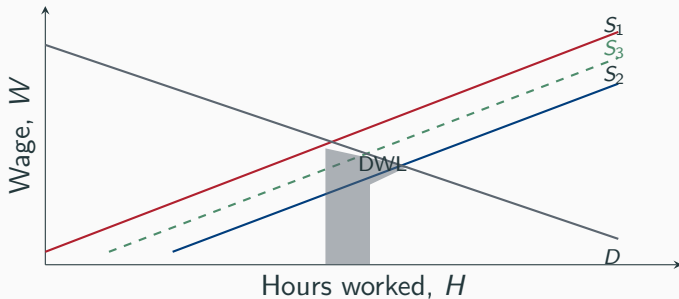
- **Consumption-lover vs. frugal person:** 74.4% say equally deserving; only 4.1% favor the consumption-lover (against pure utilitarianism, which would favor the higher-MU consumption-lover).
- **Hard-working (long hours, low wage) vs. leisure-lover (short hours, high wage):** 42.7% favor the hard-working person; only 2.9% favor the leisure-lover.
- **Ranking deservingness of a benefit increase:** disabled person ranked most deserving 57.5% of the time; a welfare recipient *not* looking for work ranked *least* deserving 70.8% of the time.

People reward effort and penalize perceived free-riding — fairness judgments do not match any single textbook SWF.

Source: Saez, E. and Stantcheva, S. (2016), "Generalized Social Marginal Welfare Weights for Optimal Tax Theory," *American Economic Review*.

2.4 Welfare Implications of Reforming Sozialhilfe

Efficiency: The Labor Market View



S_1 no Sozialhilfe; S_2 base regime (100% withdrawal); S_3 with the reform (65% withdrawal); D = labor demand.

Sozialhilfe shifts supply left ($S_1 \rightarrow S_2$); the reform shifts it partly back ($S_2 \rightarrow S_3$), shrinking the deadweight loss.

Equity: Single-Parent Poverty in Austria

- **36%** of single-parent households were at risk of poverty in 2024 — up from just **14%** in 2014.
- **43%** were at risk of poverty *or* social exclusion (the broader EU measure).
- Single parents spend **26%** of income on housing, versus **12%** for two-parent, two-child households.

Single parents — the population Sozialhilfe most directly supports — are also the population for whom benefit cuts carry the largest equity cost.

Source: Statistik Austria, EU-SILC (2024); Sozialministerium, “Armutsgefährdung und soziale Ausgrenzung von Ein-Eltern-Haushalten.”

The Trade-off in Practice

- Cutting Sozialhilfe's effective reduction rate **raises efficiency** (more labor supplied, smaller deadweight loss) — *if* the substitution effect dominates.
- It also **worsens equity** for those unable to work more (illness, caregiving, no available jobs) — exactly the single parents most at risk of poverty.
- Whichever social welfare function — or “just deserts” criterion — you find most persuasive, the trade-off does not disappear; it only shifts *where* you draw the line.

2.5 Conclusion

Highlights

- Constrained utility maximization (indifference curves + budget constraint) is the workhorse model of individual choice; profit maximization plays the same role for firms.
- Price changes split into substitution and income effects — both matter, and can point in different directions for the labor-leisure choice.
- The competitive equilibrium maximizes total surplus (1st Welfare Theorem), but says nothing about its distribution.
- The 2nd Welfare Theorem's promise of costless redistribution fails once government cannot observe the characteristics it would need to target — this *is* the equity–efficiency trade-off.
- Real people's fairness judgments (Saez–Stantcheva) don't match any single textbook social welfare function — effort and perceived deservingness matter enormously.
- Theory predicts the *direction* of Sozialhilfe reform's effects on labor supply, not their *size* — that requires the empirical tools of Lecture 3.

Discussion Questions for Next Time

1. Julia's reduction rate falls from 100% to 65%. Sketch a case (using indifference curves) where she works *less*, not more, after the reform.
2. Is Vienna's Richtwertmietzins price ceiling more like a market failure correction or a redistribution tool? Does the 1st Welfare Theorem apply?
3. Using the Saez–Stantcheva survey results, would you expect public support for Sozialhilfe to be higher or lower than support for old-age pensions? Why?
4. Pick one social justice principle (commodity egalitarianism, equality of opportunity, just deserts) and use it to argue for *or* against the 2019 Sozialhilfe reform.

Sources (1/2): Textbooks & Theory

Textbooks:

- Gruber, J. *Public Finance and Public Policy*, Ch. 2, Worth Publishers.
- Stantcheva, S. "Lecture 2: Theoretical Tools for Public Economics," Harvard EC2450A (Fall 2022).
- Becker, G. and Murphy, K. (1988), "A Rational Theory of Addiction," *Journal of Political Economy*.
- Rawls, J. *A Theory of Justice* (1971, rev. 1999), Harvard University Press.
- Saez, E. and Stantcheva, S. (2016), "Generalized Social Marginal Welfare Weights for Optimal Tax Theory," *American Economic Review*.

Sources (2/2): Austrian Data & Institutions

- Sozialministerium: Sozialhilfe-Grundsatzgesetz; “Armutsgefährdung und soziale Ausgrenzung von Ein-Eltern-Haushalten”
- ÖGB, “So viel Sozialhilfe bekommt man in Österreich”
- Statistik Austria, EU-SILC (2024) income and living conditions
- Arbeiterkammer Wien, “Richtwertmietzins”; Stadt Wien, Mietenrechner

Thank you!

Questions and discussion